

NUMERICAL METHODS FOR PDES

Lecture 2

1D Poisson with homogeneous Dirichlet
boundary conditions

$$\begin{cases} (Lu :=) -u'' = f & \Omega = (0, 1) \\ u(0) = 0 = u(1) \end{cases}$$

Centred FD

• N subintervals $h = \frac{1}{N}$

• $x_i = h i \quad i = 0, \dots, N$

• Find $U_i \approx u(x_i) \quad :$

$$-\int_h^2 U_i = F_i \quad i = 1, \dots, N-1$$

$$\left\{ \begin{array}{l} U_0 = 0 = U_N \end{array} \right.$$

where $F_i = f(x_i)$

$$\int_h U_i = -\delta_h^2 U_i = -\frac{U_{i+1} - 2U_i + U_{i-1}}{h^2}$$

In algebraic form,

$$U = \{U_i\}_{i=1}^{N-1} \quad F = \{F_i\}_{i=1}^{N-1}$$

$$A = -\frac{1}{h} \begin{bmatrix} & & & & 0 \\ & & & & \\ & & 1 & -2 & 1 \\ & 0 & & & \\ 0 & & & & \end{bmatrix}$$

$$A U = F$$

A is • tridiagonal Toeplitz

• symmetric

• diagonally dominant, positive definite

• M -matrix

Def:

M -matrix

• off-diagonal entries are non-positive

• eigenvalues have non-negative real parts

$\Rightarrow$ max principle applies

$$DMP: V = \{V_i\}_{i=0}^N, \quad \sum_{i=1}^N V_i \leq 0$$

$$\text{then } \max_{0 \leq i \leq N} V_i = \max \{V_0, V_{N+1}\} \quad \forall i=1, \dots, N-1$$

Lemma (stability):

$$\|V\|_{\infty} \leq \max\{|V_0|, |V_N|\} + \frac{1}{8} \|\sum_{i=1}^N V_i\|_{\infty}$$

proof: by using a comparison function

$$\omega(x) := x - x^2 = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2$$

$$W : W_i = \omega(x_i)$$

$$\mathcal{L}_h W_i = -\omega''(x_i) = 2$$

$$W_0 = 0 = W_H \quad 0 \leq W_i \leq \frac{1}{4}$$

$$\text{Set } \tilde{V}_i^\pm = \pm V_i - \frac{1}{2} \|\mathcal{L}_h V\|_{\ell^\infty} W_i$$

$$\mathcal{L}_h \tilde{V}_i^\pm = \pm \mathcal{L}_h V_i - \frac{1}{2} \|\mathcal{L}_h V\|_{\ell^\infty} \mathcal{L}_h W_i$$

$\uparrow$
 ≤ 0

$\parallel$
 $\cancel{2}$

$$(DMP) \Rightarrow \hat{V}_1^{\pm} \leq \{ \pm V_0, \pm V_H \}$$

$$= \pm V_1 - \frac{1}{2} \|\mathcal{L}_h V\|_{\ell_\infty} \max_i |W_i|$$

$$\Rightarrow \pm V_1 \leq \max \{ |V_0|, |V_H| \} + \frac{1}{2} \|\mathcal{L}_h V\|_{\ell_\infty} \max_i |W_i|$$

$\wedge$
 4

$\Downarrow$

uniqueness

$\Downarrow$

existence (discrete)

$\Downarrow$

well-posed

Lemma (accuracy of second central divided difference formula)

If $u \in C^4([0, 1])$, $h > 0$, for $x \in [h, 1-h]$,

error identity

$$u''(x) - \frac{u(x+h) - 2u(x) + u(x-h))}{h^2} = \frac{h^2}{24} (u^{(4)}(\xi_1) + u^{(4)}(\xi_2))$$

Boyle

with $\xi_1 \in [x-h, x]$, $\xi_2 \in [x, x+h]$

$$\begin{aligned} \frac{1}{h^2} \left(\begin{aligned} &u(x_{i+1}) - \cancel{2u(x_i)} + u(x_{i-1}) \\ &= \cancel{u(x_i)} - \cancel{h u'(x_i)} + \frac{h^2}{2} u''(x_i) \\ &\quad + \frac{h^3}{6} u^{(3)}(x_i) + \frac{h^4}{24} u^{(4)}(\xi_2) \\ &\quad - \cancel{u(x_i)} + \cancel{h u'(x_i)} + \frac{h^2}{2} u''(x_i) \\ &\quad - \frac{h^3}{6} u^{(3)}(x_i) + \frac{h^4}{24} u^{(4)}(\xi_1) \end{aligned} \right) \\ = \frac{h^2}{12} u''(x_i) + \frac{h^4}{24} (u^{(4)}(\xi_1) + u^{(4)}(\xi_2)) \end{aligned}$$

Thurs,

$$|u''(x) - \delta_h^2 u(x)| \leq \frac{h^2}{12} \|u^{(4)}\|_{L^\infty(0,1)}$$

error bound

($\Rightarrow \delta_h^2$ exact on up to cubic polynomials)

Consistency \ POE $\int u(x)$

$$T(x) = f(x) - \delta_h^2 u(x) \quad \text{Truncation error}$$

$$T_n = T(x_i)$$

Prove that $T(x) \xrightarrow{h \rightarrow 0} 0$

$$T(x) = f(x) - (-\delta_h^2 u(x))$$

$$= \cancel{f(x)} - \left[- \left(\cancel{u''(x)} + \frac{h^2}{24} \left(u^{(4)}(\xi_1) + u^{(4)}(\xi_2) \right) \right) \right]$$

$$= \frac{h^2}{24} \left(u^{(4)}(\xi_1) + u^{(4)}(\xi_2) \right)$$

$$|T(x)| \leq \frac{h^2}{12} \|u^{(4)}\|_{L^\infty(0,1)}$$

$$\text{so } T(x) \xrightarrow{h \rightarrow 0} 0 \quad \text{or } O(h^2)$$

$$\text{that is } \lim_{h \rightarrow 0} \frac{T(x)}{h^2} \text{ finite}$$

$\Rightarrow$ FD method is consistent

Theorem (error estimate in L_∞)

$\forall h > 0$, the unique FD solution

U satisfies

$$\max_i |u(x_i) - U_i| \leq \frac{h}{96} \overset{(2)}{\parallel u^{(4)} \parallel}_{L^\infty(0,1)}$$

We say that the FDM is 2nd order accurate.

Proof: $e_i := u(x_i) - U_i$ $\begin{cases} e_0 = 0 \\ e_H = 0 \end{cases}$
 $e = \{e_i\}$

① stability

$$\begin{aligned} \|e\|_{L_\infty} &\leq \max \left\{ \underbrace{\|e_0\|}_0, \underbrace{\|e_H\|}_0 \right\} + \frac{1}{8} \|\mathcal{L}_h e_i\|_{L_\infty} \\ &= \frac{1}{8} \|\mathcal{L}_h e_i\|_{L_\infty} \end{aligned}$$

② Consistency

$$\mathcal{L}_h e_i = \int_h u(x_i) - \int_h U_i$$

FDM)

$$= \int_h u(x_i) - f_i$$

$$= T_i$$

consistency

$$\Rightarrow \|e_i\|_{L^\infty} \leq \frac{1}{8} \|T_i\|_{L^\infty} \leq \frac{1}{8} \cdot \frac{h^2}{12} \|u^{(4)}\|_{L^\infty}$$

In fact we have proven:

Calling $I_h u$ interpolation

operator $(I_h u)_i = u(x_i)$

$$\|I_h u - u\|_{L^\infty} \leq \frac{\overbrace{h^2}^{\text{order}}}{\underbrace{96}_{\text{constant}}} \|u^{(4)}\|_{L^\infty}$$

$$\|u - I_h u\|$$

DIVIDED DIFFERENCES

$$\delta_h^2 u(x) = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

$$|u''(x) - \delta_h^2 u(x)| \leq \frac{h^2}{12} |u^{(4)}|_{L^\infty}$$

2nd order

$$\delta_h u(x) = \frac{u(x+h/2) - u(x-h/2)}{h}$$

or

$$\delta_{2h} u(x) = \frac{u(x+h) - u(x-h)}{2h}$$

2nd order

$$|u'(x) - \delta_{2h} u(x)| \leq \frac{h}{6} |u^{(3)}|_{L^\infty}$$

$$\delta_{h,+} u(x) = \frac{u(x+h) - u(x)}{h}$$

forward FD

$$\delta_{h,-} u(x) = \frac{u(x) - u(x-h)}{h}$$

backward

$$|u'(x) - \delta_{h, \pm} u(x)| \leq \frac{h}{2} |u^{(3)}|_{L^\infty}$$

1st order

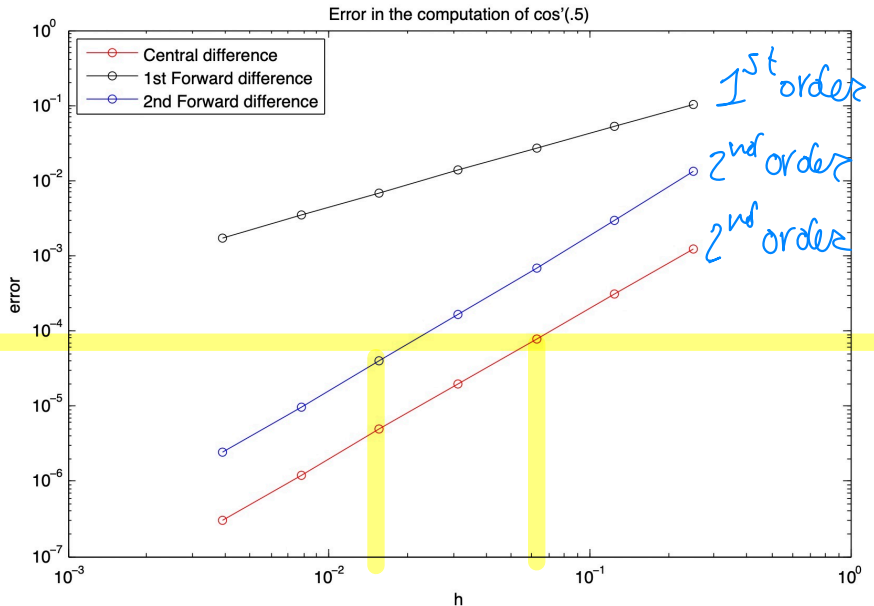
, correction

$$\delta_{h, -1/2} u(x) = - \left(\delta_{h, -} + \frac{h}{2} \delta_{h, -}^2 \right) u(x)$$

$$= \frac{1}{2h} \left(-u(x-2h) + 4u(x-h) - 3u(x) \right)$$

$$|u'(x) - \delta_{h, -1/2} u(x)| \leq h^2 |u^{(3)}|_{L^\infty}$$

See the error bounds in action :



The centered formula provides the same accuracy with bigger h !

General 2 points b. v. p.

$$\mathcal{L} u := \begin{cases} -a u'' + b u' + c u = f & \Omega = (0, 1) \\ u(0) = u_0 \\ u'(1) = v_1 \end{cases}$$

$$a > 0, \quad c \geq 0$$

$$f_i = f(x_i); \quad a_i = a(x_i); \quad b_i = b(x_i)$$

$$c_i = c(x_i)$$

$$x_i; \quad U_i \approx u(x_i) \quad \parallel \quad \frac{U_{i+1} - U_{i-1}}{2h}$$

$$\begin{cases} -a_i \delta_h^2 U_i + b_i \delta_{2h} U_i + c_i U_i = f_i \\ U_0 = u_0 \\ \frac{1}{2h} (-U_{N-2} + 4U_{N-1} - 3U_N) = v_1 \end{cases}$$

one-sided 2nd order divided difference approximating $u'(x_N) = v_1$